

Anjali Sapkota

Project Specification_Intial Project Plan _Risk Register 2025_26 77466874

 Component 1: Project Specification (Initial Project Plan + Risk Register)

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BSc (Hons) Computing Course 2025/26

Level 6 Production Project

Name: Anjali Sapkota

Student I.D.: 77466874

Course: BSc (Hons) Computing

Supervisor's Name:

Final Project Individual Aim & Objectives

Title of my Project:

DetectDerm: AI-Powered Skin Disease Identifier for Nepali Users

Aim of my Project:

This project aims to develop DetectDerm, a tool that can detect and identify 3–5 common skin diseases from images and provide Nepali-language advice for care or consultation. Using a pre-trained CNN model with public image datasets, the tool will be mobile-friendly and work offline, ensuring accessibility for users in both rural and urban areas of Nepal. The project focuses on delivering accurate, practical, and user-friendly skin disease detection.

Objectives of my Project:

- Detects and classifies 3–5 common skin diseases from images using an AI model (CNN with Transfer Learning).
- Provides Nepali-language recommendations for care or consultation for each detected condition.
- Builds a mobile-friendly, offline-first prototype for accessibility in rural and urban Nepal.
- Measures accuracy, loss, and confusion matrix to evaluate model performance.
- Applies image preprocessing and data augmentation to improve model predictions.
- Collects user feedback from 2–3 users to refine the interface and usability.
- Prepares documentation, flow diagrams, screenshots, and final report for project presentation and evaluation.

Specification of my Product:

Functional Requirements (MoSCoW)

Requirement	MoSCoW
Upload skin images for analysis	M
Detect and classify 3–5 common skin diseases	M
Provide Nepali-language recommendations	M
Display model accuracy/confidence	M
Work offline after initial setup	M
Mobile-friendly user interface	S
Allow adding new skin diseases in future	C
Integrate more than one AI model for comparison	C
Fine-tune or retrain model with large or new datasets	W
Real-time skin disease detection using video input	W
Automatic dataset collection from users	W

Non-Functional Requirements (MoSCoW)

Requirement	MoSCoW
Developed using free software tools (Python, Keras, Flask)	M
Product is tested and validated with test images	M
Simple and user-friendly interface	S
Platform-independent (works on web / mobile browsers)	C
Model prediction generated within a reasonable time (15–20 seconds)	S
Identify the most suitable AI model for Nepal	C
Advanced offline caching or full PWA support	W
Support AI frameworks other than TensorFlow/Keras	W
Enterprise-level scalability and security features	W

Research:

Studying skin diseases from images is very important for healthcare. Convolutional Neural Networks (CNNs) can analyze many skin images and learn to identify common skin conditions by recognizing patterns (Rajakumar, 2024). Using pre-trained models such as VGG16, VGG19, and MobileNet with transfer learning improves accuracy even when datasets are small (Ahmmed et al., 2025). Tools like TensorFlow and its Detection API help developers build, train, and deploy these models efficiently, using features that have already been learned (Fujisawa et al., 2021; Duhok Polytechnic University & A.M., 2024). Studies show that these methods can correctly detect over 90% of skin conditions, including psoriasis, melanoma, and actinic keratosis (Diagnostics, 2025). Although challenges like uneven datasets and diverse skin types exist, these methods provide a solid foundation for creating an offline, mobile-friendly tool that offers advice in Nepali for users in both rural and urban areas.

Evaluation:

The DetectDerm project will be checked to see if it meets all its goals. Accuracy will be tested using a set of skin images that were not used for training, including 3–5 common skin problems like psoriasis, melanoma, and actinic keratosis. The model will be measured using accuracy, loss, and confusion matrix to see how well it works. Different types of images with varied skin tones, lighting, and quality will also be tested. The expected result is the correct disease name and advice in Nepali. Feedback from 2–3 users will help check if the app is easy to use, mobile-friendly, and works offline.



7

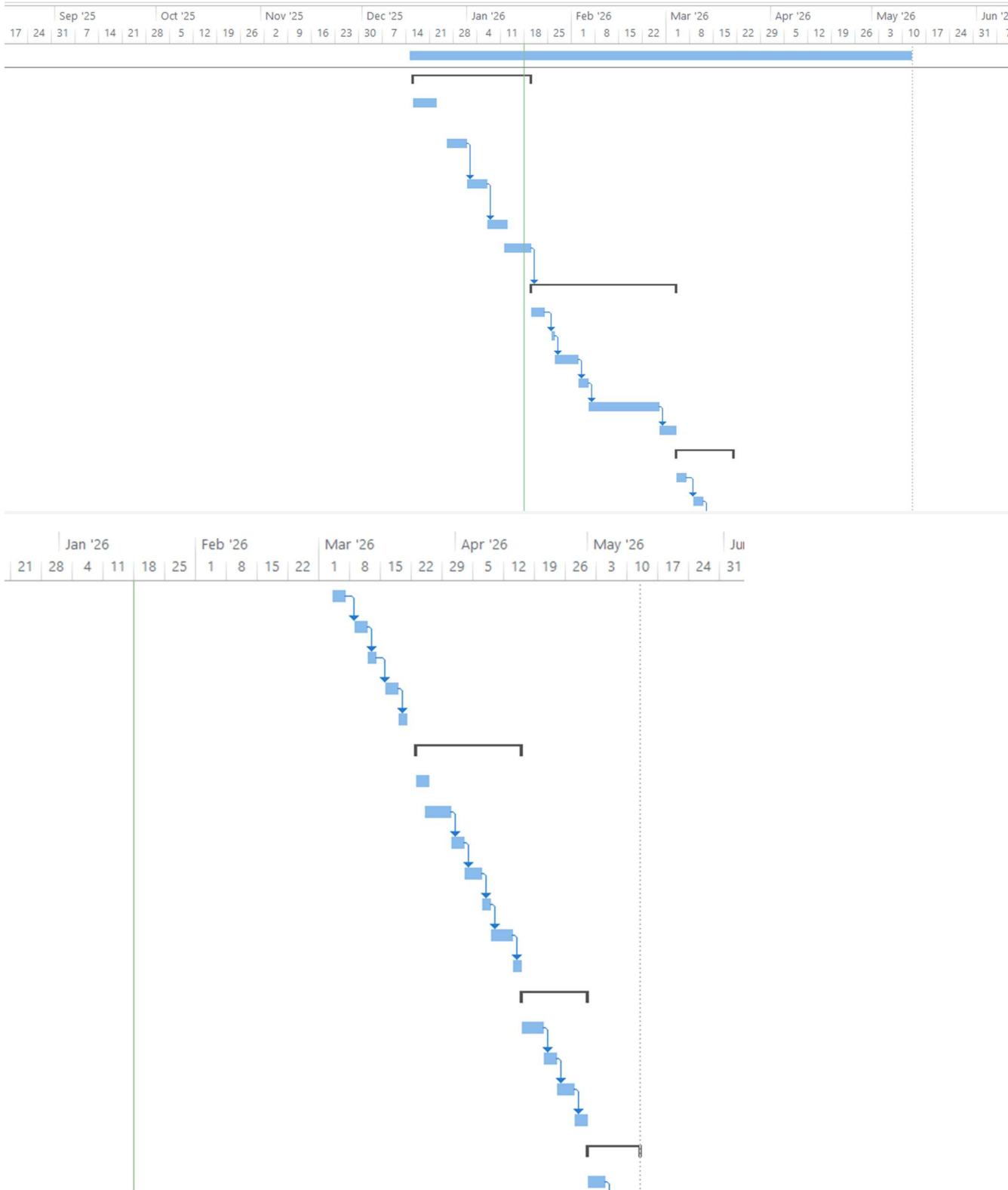
Project Planning & Methodology

Project Planning:

	Task	Task Name	Duration	Start	Finish	Predecessor
		Final Project Start	107 days?	Mon 12/15/25	Tue 5/12/26	
		Initiation	25 days	Tue 12/16/25	Mon 1/19/26	
		Research on the module requirements	5 days	Tue 12/16/25	Mon 12/22/25	
		Brainstorming on feasible project ideas	4 days	Fri 12/26/25	Wed 12/31/25	
		Research on AI-based skin disease detection	4 days	Thu 1/1/26	Tue 1/6/26	4
		Project title finalization	4 days	Wed 1/7/26	Mon 1/12/26	5
		Initial project plan preparation and submission	6 days	Mon 1/12/26	Mon 1/19/26	
		Planning	31 days	Tue 1/20/26	Tue 3/3/26	7
		Define project aim, objectives, and scope	4 days	Tue 1/20/26	Fri 1/23/26	
		Supervisor allocation	1 day	Mon 1/26/26	Mon 1/26/26	9
		Requirement analysis (functional & non-functional)	5 days	Tue 1/27/26	Mon 2/2/26	10
		Selection of development methodology (Agile)	3 days	Tue 2/3/26	Thu 2/5/26	11
		Dataset identification and feasibility study	15 days	Fri 2/6/26	Thu 2/26/26	12
		Ethical approval form preparation and submission	3 days	Fri 2/27/26	Tue 3/3/26	13
		Research on Project	13 days	Wed 3/4/26	Fri 3/20/26	

		Research on Project	13 days	Wed 3/4/26	Fri 3/20/26	
		Literature review on skin disease detection	3 days	Wed 3/4/26	Fri 3/6/26	
		Review of CNN and transfer learning techniques	3 days	Mon 3/9/26	Wed 3/11/26	16
		Study of existing systems and tools	2 days	Thu 3/12/26	Fri 3/13/26	17
		Dataset analysis and selection	3 days	Mon 3/16/26	Wed 3/18/26	18
		Literature review and methodology submission to supervisor	2 days	Thu 3/19/26	Fri 3/20/26	19
		Training and Implementation	18 days	Mon 3/23/26	Wed 4/15/26	
		Image preprocessing and data augmentation	3 days	Mon 3/23/26	Wed 3/25/26	
		CNN model selection and configuration	4 days	Wed 3/25/26	Mon 3/30/26	
		Model training using transfer learning	3 days	Tue 3/31/26	Thu 4/2/26	23
		Model evaluation (accuracy and loss)	2 days	Fri 4/3/26	Mon 4/6/26	24
		Application interface design	2 days	Tue 4/7/26	Wed 4/8/26	25
		Integration of trained model into application	3 days	Thu 4/9/26	Mon 4/13/26	26
		Offline functionality implementation	2 days	Tue 4/14/26	Wed 4/15/26	27
		Testing	11 days	Thu 4/16/26	Thu 4/30/26	
		Functional testing using test images	3 days	Thu 4/16/26	Mon 4/20/26	
		Accuracy validation and confusion matrix generation	3 days	Tue 4/21/26	Thu 4/23/26	30
		User testing with 2–3 users	2 days	Fri 4/24/26	Mon 4/27/26	31
		Bug fixing and performance improvement	3 days	Tue 4/28/26	Thu 4/30/26	32
		Closing	8 days	Fri 5/1/26	Tue 5/12/26	
		Final system evaluation	2 days	Fri 5/1/26	Mon 5/4/26	
		Documentation preparation (report, diagrams, screenshots)	2 days	Tue 5/5/26	Wed 5/6/26	35
		Product presentation preparation	1 day	Thu 5/7/26	Thu 5/7/26	36
		System demonstration	2 days	Fri 5/8/26	Mon 5/11/26	37
		Final report submission	2 days	Mon 5/11/26	Tue 5/12/26	
		Meetings with Supervisor	75 days?	Mon 1/26/26	Fri 5/8/26	

Figure: Task sheet of my project



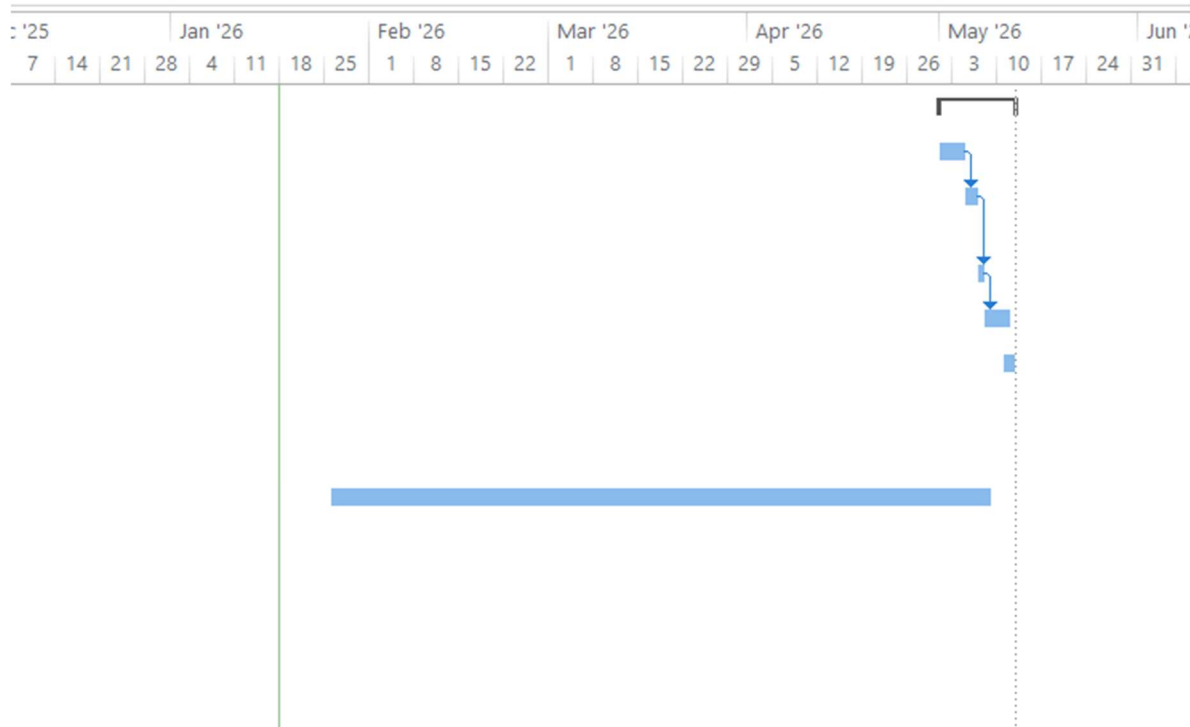


Figure of Gantt chart

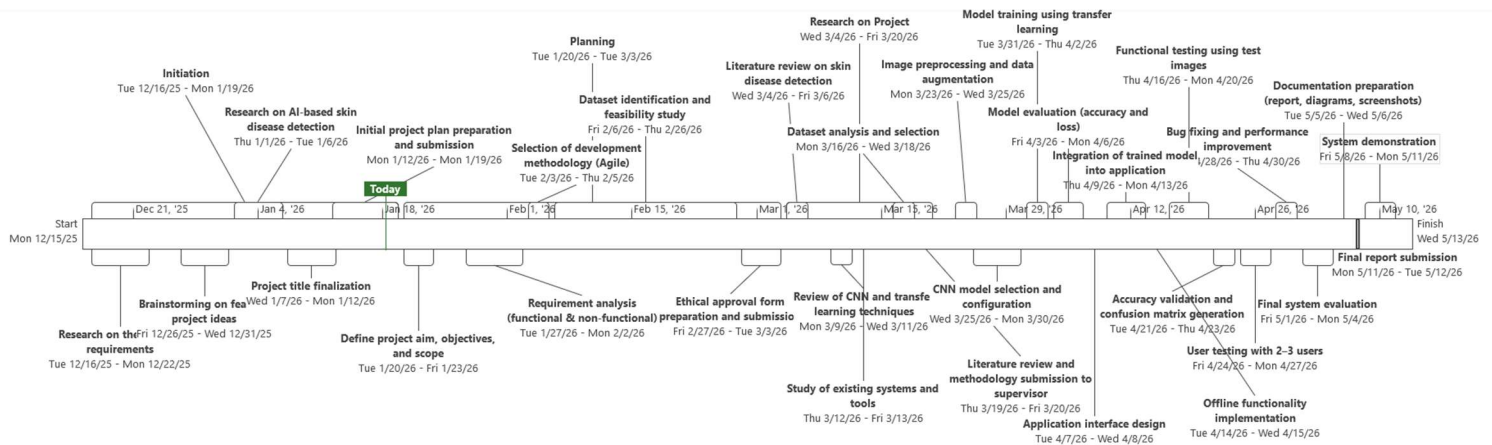


Figure of Timeline

Methodology:

For DetectDerm, the Agile methodology is chosen because it allows flexible and iterative work, which is ideal for a 12-week project. Agile enables development in short cycles, focusing on tasks like collecting skin images, training CNN models, evaluating results, and building the mobile interface. It also allows user feedback to be included quickly, improving usability and accuracy for Nepali

users. Other approaches, such as Waterfall and RAD, were not selected. Waterfall is too rigid and does not support changes during model training, while RAD lacks structured sprints for systematic testing and improvement. Previous studies support this approach: Rajakumar (2024) and Ahmmed et al. (2025) show that iterative training improves CNN accuracy, Fujisawa et al. (2021) emphasizes testing in practical environments, and Diagnostics (2025) and Duhok Polytechnic University & A.M. (2024) highlight the benefits of flexible, iterative methods for AI projects. Therefore, Agile is the most suitable methodology for efficiently developing DetectDerm.

Resources

The hardware and software I require to complete my Project successfully:

List of Hardware

- Laptop
- Mobile

List of Software

- Python 3.x
- TensorFlow / Keras
- OpenCV
- NumPy, Pandas, Matplotlib
- Flutter or React Native
- Flask
- Jupyter Notebook or VS Code
- Chrome browser
- Git / GitHub
- Documentation, flow diagrams, screenshots

Human Resource

I am working on my Project with the following people

Name: Anjali Sapkota

Role:

Module Leader: Rohit Raj
Pandey

Supervisor:

Initial Bibliography

1. Rajakumar, A., Priya, D. & Muthulakshmi, R., 2024. *Skin Disease Classification and Detection Using Deep Learning*. *Frontiers in Health Informatics*, 13(3). Available at: <https://pubmed.ncbi.nlm.nih.gov/36655135> [Accessed 18 January 2026].
2. Ahmmed, F., Raihan, M.Z., Nahar, K., Asadujjaman, D.M. & Rahman, M.M., 2025. *Skin Disease Detection and Classification of Actinic Keratosis and Psoriasis Utilizing Deep Transfer Learning*. *arXiv*, January. Available at: <https://arxiv.org/abs/2501.13713> [Accessed 18 January 2026].
3. Fujisawa, Y., et al., 2021. *A Deep Learning Based Framework for Diagnosing Multiple Skin Diseases in a Clinical Environment*. *Frontiers in Medicine*. Available at: <https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2021.626369/full> [Accessed 18 January 2026].
4. Diagnostics, 2025. *Skin Lesion Classification and Detection Using Machine Learning Techniques: A Systematic Review*. *Diagnostics*. Available at: <https://www.mdpi.com/2075-4418/13/19/3147> [Accessed 18 January 2026].
5. Duhok Polytechnic University & A.M., 2024. *Skin Disease Recognition Based on Deep Learning Algorithms: A Review*. *The Indonesian Journal of Computer Science (IJCS)*, 13(3). Available at: <https://ijcs.net/ijcs/index.php/ijcs/article/view/3991> [Accessed 18 January 2026].

Risk and Register

ID	Risk	Risk Description	Likelihood	Impact	Severity	Owner	Mitigation	Status
1	Dataset issue	Enough good quality skin images may not be available	Medium	High	High	Anjali	Use different public datasets and apply data augmentation	Open
2	Low model accuracy	The model may not give accurate results	Medium	High	High	Anjali	Use transfer learning and test with different settings	Open
3	Overfitting	Model may work well only on training images	Medium	Medium	Medium	Anjali	Use validation data and data augmentation	Open
4	Hardware limitation	Laptop may be slow during training	Medium	Medium	Medium	Anjali	Use lightweight models and reduce image size	Open
5	Time shortage	Tasks may take more time than planned	Low	High	Medium	Anjali	Follow weekly plan and meet supervisor regularly	Open

6	Offline feature problem	Offline mode may not work properly	Low	Medium	Low	Anjali	Keep offline features simple and test early	Open
7	UI difficulty	Users may find the app hard to use	Low	Medium	Low	Anjali	Take feedback from 2–3 users and improve design	Open
8	Ethical approval delay	Ethical approval process may be slow	Low	Medium	Low	Anjali	Submit ethics form early	Open
9	Software bugs	Errors may appear during development	Medium	Medium	Medium	Anjali	Test the system during development and fix bugs	Open
10	Data privacy concern	Users may worry about image safety	Low	High	Medium	Anjali	Process images locally and do not store them	Open
11	Laptop failure	Laptop may stop working during the project	Low	High	Medium	Anjali	Backup files on GitHub/USB and use lab if needed	Open
12	Insufficient research	Not enough suitable research	Medium	Medium	Medium	Anjali	Use Google Scholar and discuss	Open

		papers may be found					with superviso r	
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